

Lecture 7 ERS312 Notes

3rd Topic Ocean Floor Topography and Plate Tectonics

-bottom of the ocean is flat and featureless

What were findings of the publication of the first map of the Atlantic Ocean seafloor (that there is a huge underwater mountain chain, was a rift in the center *narrow elongated valley)

*people started speculating about plate tectonics

What is an important feature of an active continental margin coast to deep-sea cross-section: Ocean trench is a defining feature of an active continental margin

Three characteristics of the Pacific Ring of Fire:

-volcanoes, earthquakes, and trenches all around

Why were continental shelf areas exposed during the ice ages?

-during ice ages was cold so precipitation on lands falls as snow and turned into ice

What is depth of shelf break and why is it uniform across the world?

-depth of shelf break is around 120m (it is uniform as most erosion happens at sea level)

What drives the ice age

-the orbit of the earth and the fact that the earth's tilt is slowly changing, and orientation of earth axis is changing (3 main parameters)

Which important landmasses were connected during the Pleistocene

-Asia and North America, Russia and Alaska, Asia and Europe

Turbidity Currents

- these currents travel down the slope at fast speeds
- turbulence of water increases at bottom of slope

Abyssal Plains

- fully flat
- covered with deep sea sediments
- abyssal hills are less than 1km above sea floor (everything higher is sea mounts)
- seamounts are extinct volcanoes (that never reached above sea-level)
- table mounts are extinct volcanoes (have once reached above sea level)
- Hawaii islands built by volcanoes
- Hawaiian seamounts were once active volcanoes that are now inactive

Ocean Trenches

- find them around pacific ring of fire
- Indian ocean and Atlantic Ocean have smaller trenches
- steep v shaped depressions on ocean floor
- ocean trenches almost always extend along continents/islands
- examples: Peru-Chile trench

Mid-Ocean Ridges

- were underwater mountain chains found
- when/where was most recent eruption on Mid-Ocean ridge (in Iceland)
- mid-ocean ridge in Iceland is so shallow that its above sea level

Continental Drift Theory

- plate tectonics only developed after 1950s but there was an early proponent of plate tectonics proposed in 1912

